FiremeX
Fire & Security Monitoring

CCTV-BASED FIRE DETECTION SYSTEM

TEST PLAN

Master Test Plan for CCFDS Software Release 1.0

Document ID	QA-01
Project	CCTV-Based AI Fire Detection System
Prepared By	Jayashini Jayaweera
Date	Jul-2026
Status	Approved for Release Cycle 1

1. Introduction

This Test Plan defines the scope, approach, resources, and schedule for testing the CCTV-Based Fire Detection System. This system ingests live RTSP/RTMP video feeds from CCTV cameras, applies an AI/ML-based fire and smoke detection model, and raises real-time alerts through a central monitoring dashboard, SMS/email notifications.

2. Objectives

- Verify fire and smoke detection accuracy under varied lighting, weather, and camera conditions.
- Validate real-time alert generation and notification delivery within defined SLA (≤ 5 seconds).
- Confirm integration with camera feeds (ONVIF/RTSP), dashboard UI.
- Ensure system reliability for 24x7 continuous monitoring with minimal false positives/negatives.
- Validate data storage, audit logging, and historical incident retrieval.
- The system uses historical alerts and incident records to identify recurring patterns and continuously improve the AI model, reducing false alarms and increasing detection accuracy for frequently occurring scenarios.

3. Scope

3.1 In Scope

- Functional testing of video ingestion, detection engine, alerting, and dashboard modules.
- Non-functional testing: performance, load, security, and reliability of the detection pipeline.
- Integration testing with CCTV/NVR systems, notification gateways (SMS/Call).
- Regression testing for every AI model version update.
- Usability testing of the monitoring dashboard for control-room operators.

3.2 Out of Scope

- Testing of third-party fire-suppression hardware actuation (mechanical).
- Detailed AI model training pipeline validation (covered under Data Science validation plan).

4. Test Approach / Methodology

Test Level	Description	Owner
Unit Testing	Developer-level testing of detection algorithm modules and APIs	QA Team
Integration Testing	Camera feed ingestion, AI engine, alert & notification module integration	QA Team
System Testing	End-to-end validation of fire detection to alert delivery flow	QA Team
Performance Testing	Multi-camera concurrent stream load and detection latency testing	QA Team
Security Testing	Access control, API vulnerability testing	QA Team
Regression Testing	Re-validation after every model/software build update	QA Team

5. Test Environment

- Staging environment mirroring production: IP CCTV cameras (indoor/outdoor), NVR, GPU inference server.
- Simulated fire/smoke video datasets (day, night, fog, rain) for regression suites.
- Test dashboard instance connected to a sandbox SMS/Call gateway and mock fire-panel API.

6. Entry Criteria

- Build deployed successfully to QA/staging environment.
- Unit testing completed with ≥ 90% code coverage on core detection modules.
- Test data sets (fire/smoke/no-fire video clips) prepared and validated.
- Test cases reviewed and approved.

7. Exit Criteria

- 100% of planned test cases executed; ≥ 95% pass rate.
- No open Critical or High severity defects.
- Detection accuracy ≥ agreed threshold (e.g., 95% true positive, ≤2% false positive on validation set).
- Performance and security sign-off obtained.

8. Suspension & Resumption Criteria

Testing will be suspended if the detection engine fails to start, camera feed ingestion is completely broken, or a Critical defect blocks more than 30% of planned test cases. Testing resumes once a fix is deployed and smoke-tested.

9. Roles and Responsibilities

Role	Responsibility
QA Lead	Test planning, strategy, sign-off, stakeholder reporting
Test Engineers	Test case design, execution, defect logging
Automation Engineer	Regression automation suite for detection & UI
Performance Engineer	Load/stress testing of video pipeline
Business Analyst	Requirement clarification, UAT coordination
Developers	Defect fixes, unit testing, environment support

10. Test Deliverables

- Test Plan, Test Cases, Traceability Matrix
- Test Execution Report, Defect Reports
- Performance & Security Test Reports
- Test Closure Report

11. Schedule (High Level)

Phase	Start	End
Test Planning & Design	01-Jul-2026	10-Jul-2026
Test Environment Setup	08-Jul-2026	12-Jul-2026
Functional & Integration Testing	13-Jul-2026	28-Jul-2026
Performance & Security Testing	29-Jul-2026	05-Aug-2026
UAT	06-Aug-2026	10-Aug-2026
Test Closure & Sign-off	11-Aug-2026	12-Aug-2026

12. Risks and Mitigation

Risk	Mitigation
Insufficient real fire/smoke video samples for testing	Use synthetic/augmented datasets and controlled test burns from vendor labs
High false-positive rate impacting reliability	Threshold tuning cycles with QA feedback before each release
Network latency affecting real-time alerting	Dedicated performance test on production-like network conditions
Camera hardware variability (brand/resolution)	Test matrix covering multiple camera makes/models supported

13. Approval

This Test Plan requires sign-off from: QA Lead, Project Manager, and Product Owner prior to test execution start.